SOLUTION
SECTION-I

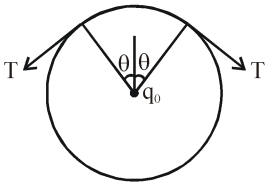
Single Correct Answer Type **10 Q. [3 M (–1)]**

1. Ans. (A)

Sol. More the charge more electric field lines density.
Electric field lines generates from +ve charge and meet on –ve charge.

2. Ans. (D)

Sol.



θ is small, then

$$2T\sin\theta = kq_0\left(\frac{Q}{2\pi R}\right)\frac{(R2\theta)}{R^2}$$

Putting $\sin\theta = \theta$ we get

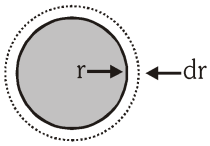
$$T = \frac{1}{8\pi^2\epsilon_0}\frac{Qq_0}{R^2}$$

$$\text{Now } Y = \frac{TR}{A\Delta R}$$

$$\therefore\ \Delta R = \frac{TR}{AY} = \left(\frac{1}{8\pi^2\epsilon_0}\frac{Qq_0}{R}\right)\left(\frac{1}{AY}\right)$$

3. Ans. (A)

Sol. Charge in layer of width dr



$dq = 2\pi r dr\sigma$
Work to bring dq = dU=Vdq
$dU = 4\sigma r\cdot 2\pi r dr\sigma = 8\pi\sigma^2 r^2 dr$

$$U = 8\pi\sigma^2\int_0^R r^2 dr = \frac{8\pi\sigma^2}{3}R^3$$

4. Ans. (A)

Sol. $mv = 2m(v')$

$$\frac{KQ(-Q)}{5R} = \frac{KQ(-Q)}{2R} + \frac{1}{2}mv^2 + \frac{1}{2}(2m)(v')^2 \Rightarrow v = \sqrt{\frac{2KQ^2}{5mR}}$$

5. Ans. (A)

Sol.
$$V_{(x,y)} = \frac{kq}{\left[(x-b)^2+y^2\right]^{1/2}} + \frac{k\left(-\frac{cq}{b}\right)}{\left[\left(x-\frac{c^2}{b}\right)^2+y^2\right]^{1/2}} = 0$$

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$$\frac{kq}{\left[(x-b)^2+y^2\right]^{1/2}}=\frac{c}{b}\frac{kq}{\left[\left(x-\frac{c^2}{b}\right)^2+y^2\right]^{1/2}}$$

$$b^2\left[\left(x-\frac{c^2}{b}\right)^2+y^2\right]=c^2\left[(x-b)^2+y^2\right]$$

$$b^2\left[x^2+\frac{c^4}{b^2}-\frac{2c^2}{b}x+y^2\right]=c^2[x^2+b^2-2bx+y^2]$$

$(b^2-c^2)\,x^2+(b^2-c^2)\,y^2-2c^2bx+2c^2bx+c^4-c^2b^2=0$

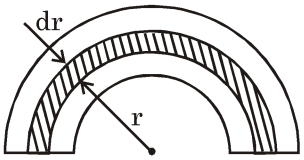
$(b^2-c^2)\,x^2+(b^2-c^2)\,y^2-c^2(b^2-c^2)=0$

$x^2+y^2=c^2 \rightarrow$ circle

6. Ans. (A)

7. Ans. (D)

Sol.



$dq=\sigma 2\pi r dr$

$$\lambda=\frac{dq}{\pi r}=2\sigma dr$$

$$E=\int_a^b\frac{2k\lambda}{r}$$

8. Ans. (B)

Sol. Charge on sphere $Q=\int_0^R\rho_0\left(4\pi r^2dr\right)=\pi\rho_0R^4$

By definition of potential $V_{2R}-V_\infty=\frac{kQ}{2R}\Rightarrow V_\infty=-\frac{kQ}{2R}=-\frac{Q}{8\pi\epsilon_0 R}=-\frac{\rho_0R^3}{8\epsilon_0}$

9. Ans. (C)

Sol. $V=-\int\vec{E}\cdot d\vec{r}$

$$V=-\int\left(\sqrt{3}\,\hat{i}-\hat{j}\right)\cdot\left(dx\,\hat{i}+dy\,\hat{j}\right)$$

$$V=-\left[\sqrt{3}x-y\right]$$

$V=-\sqrt{3}x+y$

So equipotential line having slope 60°

10. Ans. (D)

Sol. $$\vec{E}=-\frac{dv}{dx}\hat{i}-\frac{dv}{dy}\hat{j}=-2x\,\hat{i}+2y\,\hat{j}$$

$$\tan\theta=-\frac{dy}{dx}=-\frac{y}{x}\Rightarrow\frac{dy}{y}=-\frac{dx}{x}$$

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$\ell ny = -\ell nx + \ell nC$ $\quad \ell n(xy) = \ell nC$

$xy = C$

shape will be rectangular hyperbola

Multiple Correct Answer Type **4 Q. [4 M (–1)]**

11. Ans. (A,B,C,D)

Sol. $\vec{E} = (10 - 5x)i$

$m = 5kg$

$\vec{F} = q\vec{E}$

$\vec{F} = (10 - 5x)i$

(a) $F \propto -x$

so motion of charge particle is oscillatory

(b) $F = 0$ at $x = 2$

oscillation about $x = 2$

so maximum displacement is 4

$F = 10 - 5x$

$a = \dfrac{10 - 5x}{2}$

$a = 2 - x$

$a = -(x - 2)$

so $w^2 = 1$

$w = 1$

(c) $v_{max} = Aw$

$A = 2$

$w = 1$

$v_{max} = 2$

(d) velocity at mean position

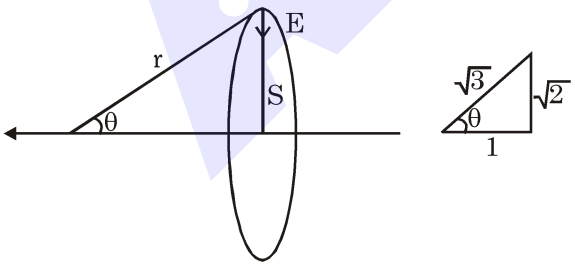
at $x = 2$

12. Ans. (B,D)

Sol. All charges are identical potential can not be zero at any point. At origin field cancels and for point on Z axis the E_z survives.

13. Ans. (B, D)

Sol.



For dipole $\tan\alpha = \dfrac{1\tan\theta}{2}$ where α is angle of electric field with position vector

$\tan\theta = \sqrt{2}$

$\dfrac{S}{r} = \sin\theta$

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$$qE = \frac{mv^2}{R}$$

$$\frac{qkp}{mr^3}\sqrt{1+3\cos^2\theta} = \frac{v^2}{s}$$

$$r = \frac{s}{\sin\theta}, \cos\theta = \frac{1}{\sqrt{3}}; \text{since} = \frac{\sqrt{2}}{\sin\theta}$$

$$\frac{kpq}{ms^2}\frac{2\sqrt{2}}{3\sqrt{3}}\left(\sqrt{2}\right) = \frac{v^2}{s}$$

$$\frac{pq}{\pi \in_0 m3\sqrt{3}} = v^2 \Rightarrow \frac{1}{s}\sqrt{\frac{pq}{\pi \in_0 m3\sqrt{3}}}$$

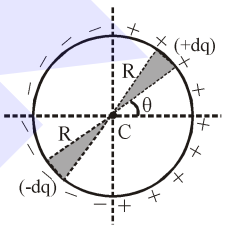
14. Ans. (A,C)

Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

15. Ans. (B)

Sol. dP = (dq)2R



$$P_{Net} = \int (dp)\cos\theta = \int (dq)2R\cos\theta$$

$$= \int (\lambda R d\theta)2R\cos\theta = 2R^2\lambda \int_{(-\pi/2)}^{(+\pi/2)} \cos\theta d\theta = 4R^2\lambda$$

16. Ans. (C)

Sol. Conservation of energy $\Delta U_{loss} = \Delta K_{gain}$

$$PE_0\left(\cos 90° - \cos 180°\right) = \frac{1}{2}I\omega^2$$

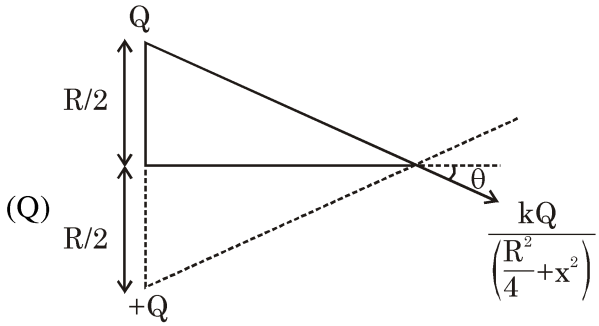
$$4R^2\lambda E_0 = \frac{1}{2}\left(mR^2\right)\omega^2 \Rightarrow \omega = \sqrt{\frac{8\lambda E_0}{m}}$$

Matching List Type (4 × 4) 1Q.[3 M (–1)]

17. Ans. (B)

Sol. (P) $E = \frac{kQx}{\left(R^2+x^2\right)^{3/2}}$ it has maximum magnitude at $x = \frac{R}{\sqrt{2}}$

$$x = \frac{-R}{\sqrt{2}}$$

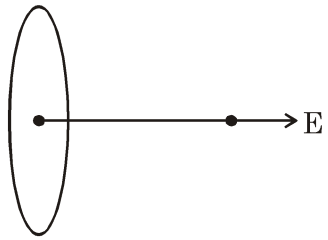


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Net electric field $= 2\left(\frac{kQ}{\left(\frac{R^2}{4}+x^2\right)}\right)\left(\frac{x}{\left(\frac{R^2}{4}+x^2\right)^{1/2}}\right)$

$= \frac{2kQx}{\left(\frac{R^2}{4}+x^2\right)^{3/2}}$

expression is same for P and Q so the graph

(R) 

$E = k\sigma 2\pi\left[1-\frac{z}{\sqrt{z^2+R^2}}\right]$

graph of it look like in option (4)

(S) Inside sphere $E \propto r$ and outside sphere $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$

So (3rd) is correct graph

SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

18. Ans. 81

Sol. $\frac{2kQq}{L} = \frac{2\times 9\times 10^9\times 3\times 10^{-3}\times 5\times 10^{-6}}{R}\times\frac{3}{5} = 81$

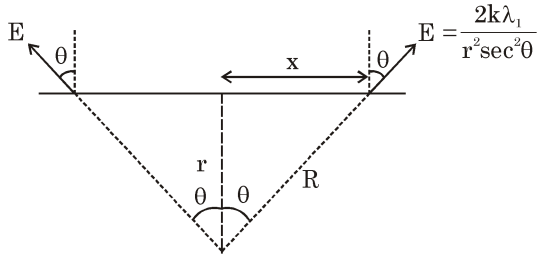
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **9 Q. [4 M (0)]**

19. Ans. 0

Sol. Electric field due to 1 at 2

$E = \frac{2k\lambda_1}{r}$ and it is radially away from z-axis.

So 

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$r \sec\theta = R$

$r \tan\theta = x$

$dx = r \sec^2\theta \, d\theta$

So component of force along y-axis added while along x force is zero.

So force = $dF = (\lambda_2 dx) E\cos\theta = \left(\frac{2k\lambda_1}{r\sec\theta}\right)\text{cosq}\ (\lambda_2\, r \sec^2\theta\, d\theta)$

$= 2k\lambda_1\lambda_2$

$$F_{net} = \int_{-\pi/2}^{\pi/2} 2k\lambda_1\lambda_2 d\theta = \frac{\lambda_1\lambda_2}{2\varepsilon_0}$$

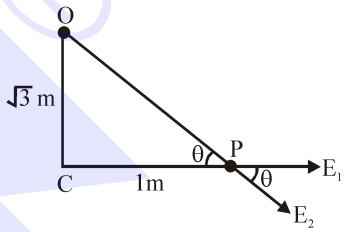
So n = 0

20. Ans. 9

Sol. $\vec{E}_r = \vec{E}_1 + \vec{E}_2$

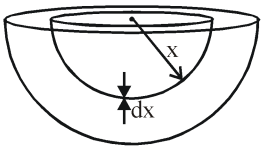
$$E_r = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos\theta}$$

where $E_1 = \frac{kQ}{12}$, $E_2 = \frac{kQ}{\left(\sqrt{3}\right)^2}$ & $\theta = 60°$



21. Ans. 3

Sol.



$$V = \int_a^b \frac{K\left(2\pi x^2 dx\right)\rho}{x}$$

$$V = 2\pi k\rho\left[\frac{b^2 - a^2}{2}\right] = \pi\rho k\,(b^2 - a^2)$$

$b = 2a$

$v = 3\pi k\rho b^2$

22. Ans. 1

Sol. Potential due to one dipole

$$V = \frac{kPi\cdot\left(\cos 30\hat{i} + \sin 30\hat{j}\right)}{\left(\frac{a}{\sqrt{3}}\right)^2}$$; so total potential = 3V

23. Ans. 5

Sol. For v_{min} to reach the origin it must cross the point x = 2L.

$\therefore$ By work energytheorem

$\Delta KE = W$ (or area)

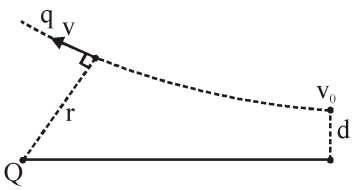
$$0 - \frac{1}{2}mv^2 = -\frac{1}{2}\times L\times E$$

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$v = \sqrt{\frac{LE_0}{m}}$ $\quad\quad$ $v = 5$ m/s

24. Ans. 2

Sol. Due to repulsion forces the point of q will be a line curved away from Q.



at least distance, r_1 we can conserve angular momentum & energy of system.

$mv_0d = mvr$

$$\& \quad 0+\frac{1}{2}mv_0^2 = \frac{KqQ}{r}+\frac{1}{2}mv^2$$

$$\frac{1}{2}mv_0^2 = \frac{KqQ}{r}+\frac{1}{2}m\left(\frac{v_0d}{r}\right)^2$$

$$\frac{1}{2}mv_0^2\left(1-\frac{d^2}{r^2}\right)=\frac{KqQ}{r}$$

$$\frac{1}{2}mv_0^2\left(r^2-d^2\right)=KqQr$$

$$\left(\frac{1}{2}mv_0^2\right)r^2-(KqQ)r-\frac{1}{2}mv_0^2d^2=0$$

$$r=\frac{KqQ+\sqrt{(KqQ)^2+m^2v_0^4d^2}}{mv_0^2}=2m$$

$KqQ = 9$

$mv_0^2 = 12$

$d = 1$

25. Ans. 2

Sol. By energy conservation

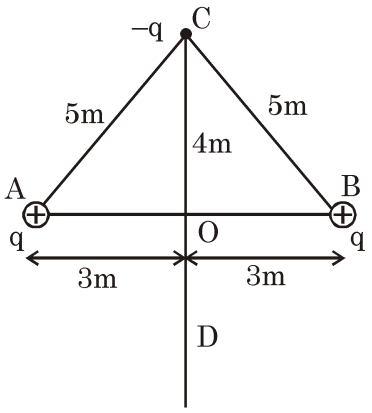
$k_1 + pE_i = kE_f + PE_f$

$$0+\left(-\frac{kq^2}{5}\right)\times 2=\frac{1}{2}mv^2+\left(-\frac{kqL}{3}\right)q$$

$$\frac{1}{2}mv^2=2\frac{kq^2}{5}\left(\frac{1}{3}-\frac{1}{5}\right)$$

$$\frac{1}{2}mv^2=\frac{4kq^2}{15}$$

$$v=\sqrt{\frac{8kq^2}{15m}}=q\sqrt{\frac{8}{4\pi\in_0\times 15m}}$$



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$$v = q\sqrt{\frac{2}{15\pi \in_0 m}}$$

$x = 2$

26. Ans. 6

Sol. $V_{(0,0,0)} = 0$ and $V_{(0,0,0.1)} = \frac{a(2)}{100} = \frac{a}{50}$

$$W_{elect} = q(V_1 - V_2) \Rightarrow 50 \times 10^{-6} = 4 \times 10^{-6} \times \frac{a}{50}$$

$\Rightarrow a = 625$

$6250 \text{ volts} = 625\,(x^2) + 625\,(y^2) + 625\,(2)\,(2)^2$

$\Rightarrow 6 = x^2 + y^2$ Thus $R = \sqrt{6}$

27. Ans. 7

Sol. $\vec{E} = \frac{a\hat{r}}{r^2}$; $V = \frac{-a}{r}$; $\Delta V = \frac{a}{r_2} - \frac{a}{r_1} = \frac{a}{5} - \frac{a}{7} = \frac{2a}{35} = 2 \times \frac{122.5}{35} = 7\text{ V}$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$|\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$

Subjective Type

2 Q. [4 M (0)]

28. Ans. $4\,py\ (y^2 + 4x^2)^{1/2}$

Sol. In non uniform electric field force on dipole

$$F = \vec{p}.\frac{d\vec{\in}}{dr}$$

$\vec{E} = 4xy^2\hat{i} + 4x^2y\hat{j}$

$\frac{d\in}{dx} = 4y^2\hat{i} + 8xy\hat{j}$

$\frac{d\in}{dy} = 8y\hat{i} + 4x^2\hat{j}$

$F_x = \vec{p} \cdot \frac{d\vec{\in}}{dx} = 4py^2$

$F_y = \vec{p} \cdot \frac{d\vec{\in}}{dy} = 8xyp$

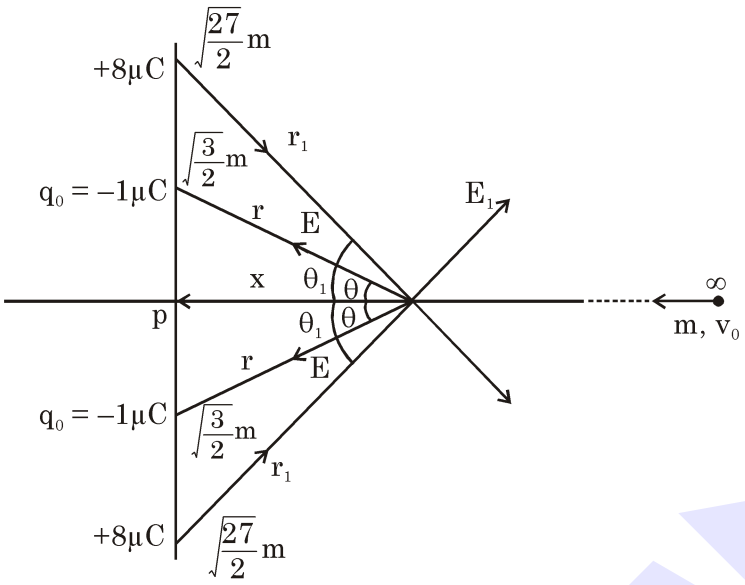
$F_{Net} = \sqrt{F_x^2 + F_y^2} = \sqrt{16p^2y^4 + 64p^2x^2y^2}$

$= 4py\left(y^2 + 4x^2\right)^{\frac{1}{2}}$

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29. **Ans.** $v_0 = 3$ m/s ; K.E. at the origin $= \left(27-10\sqrt{6}\right) \times 10^{-4}$ J approx. 2.5×10^{-4} J

Sol.



P is origin. We can see from diagram electric field at P is zero but we have to check if it is zero at any other point x-axis we have to give particle just as much velocity to reach to point where electric field is zero. (Called neutral point)

Let field is zero at θ.

$(E_\theta)_{Net} = 0$

$2E\cos\theta = 2E_0\cos\theta_1$

$$2 \times \frac{kq_0}{r^2} \times \frac{x}{r} = 2 \times \frac{k8q_0}{r_1^2} \times \frac{x}{r_1}$$

$$\frac{1}{r^3} = \frac{8}{r_1^3} \Rightarrow \frac{r_1}{r} = 2$$

For Q $\frac{r_1}{r} = +2$

$r_1 = 2r$ again $r_1 = 4$

$$\sqrt{x^2 + \frac{27}{2}} = 2\sqrt{x^2 + \frac{3}{2}} \qquad r = 2$$

$$x^2 + \frac{27}{2} = 4x^2 + \frac{12}{2}$$

$$3x^2 = \frac{15}{2} \Rightarrow x = \sqrt{\frac{5}{2}}$$

To reach at origin particle has to reach at Q the it will be attracted towards origin by electric fields of $-1\mu C$

Apply work energy theorem from ∞ to Q.

$-q\Delta V = k_f - k_i$

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$$-q\left(v_Q - v_\infty\right) = 0 - \frac{1}{2}mv_0^2$$

$$-1\times10^{-6}\left[\left(\frac{2\times8\times10^{-6}\times9\times10^{9}}{r_1} - \frac{2\times10^{-6}\times9\times10^{9}}{r}\right) - 0\right] = -\frac{1}{2}mv_0^2$$

$$-10^{6}\times2\times10^{-6}\times9\times109\left(\frac{8}{4} - \frac{1}{2}\right) = -\frac{1}{2}mv_0^2$$

Solving we get $v_0 = 3$ m/s

To find kinetic energy at P apply work energy theorem from Q $\rightarrow$ P

$-q\,(v_P - v_Q) = KE_P - KE_Q$.

$$v_P = \frac{k\left(-1\times10^{-6}\right)}{\frac{\sqrt{3}}{2}}\times2 + \frac{k\left(8\times10^{-6}\right)}{\frac{\sqrt{27}}{2}}\times2$$

$$v_Q = \frac{k\left(-1\times10^{-6}\right)\times2}{2} + \frac{k\left(8\times10^{-6}\right)}{4}\times2$$

$-q\,(v_P - v_Q) = k_P - E_Q$

$-1\times10^{-6}\,(v_P - V_Q) = k_P - 0$

Put value of v_P & v_Q and find KE at P.

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